Lecture 4: Real-Coded Genetic Algorithm (RGA)

Course: Evolutionary Algorithms

Instructor: Engr. Muhammad Farooq

Learning outcomes

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- perform one full generation of RGA by hand on a small example
- explain survivor selection using the $(\mu + \lambda)$ strategy

Connection with previous lectures

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Today's question: What should we do when the decision variables are naturally real numbers, such as lengths, voltages, gains, temperatures, or coordinates?

Why binary coding can be problematic in continuous search

Issue 1: Binary GA makes the search space discrete

A continuous variable is forced into a finite number of binary levels.

- The search is no longer truly continuous.

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For many engineering problems, the original variables are already real valued. Encoding them into bits is often unnecessary extra work.

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1. Hamming cliff

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- Here l is the number of bits used for variable x_i.
- With fixed l, the precision is fixed.
- To get finer precision, we must increase the string length.

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Important point: Selection remains conceptually the same because it only needs fitness values. The main changes happen in crossover and mutation.

Generalized framework of evolutionary computation

Generic algorithm

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In RGA, the framework stays familiar. The major redesign is in the **representation** and the **variation operators**.

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6. end while

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Real-parameter optimization using EC techniques

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RGA is the closest extension of binary GA. It keeps the same overall genetic framework, but operates directly on real-valued variables.

Worked example: Rosenbrock function

We explain the working of RGA through the Rosenbrock function:

$$f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

with bounds

$$-5 \leq x_1 \leq 5, \qquad -5 \leq x_2 \leq 5$$

- Objective: minimize $f(x_1, x_2)$

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- Objective: minimize $f(x_1, x_2)$
- Global optimum: $\boldsymbol{x}^* = (1, 1)^T$
- Optimum value: $f(\boldsymbol{x}^*) = 0$

Why Rosenbrock is a useful teaching example

Why this function is important

- continuous search space

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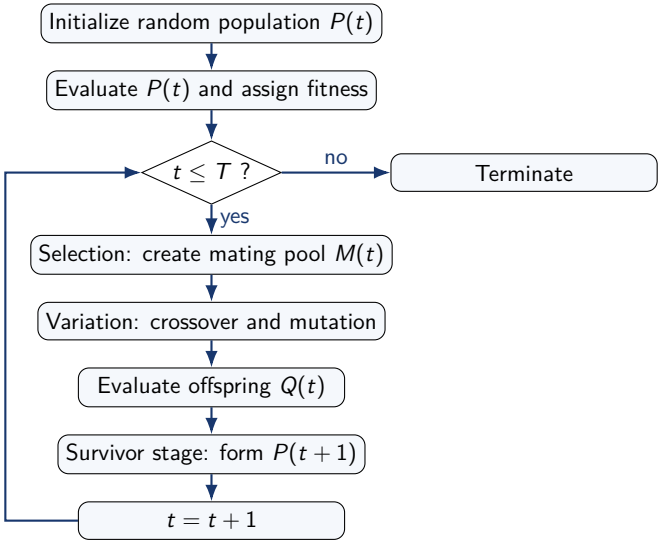
A good optimizer must not only improve the objective value, but also move carefully inside a sensitive continuous landscape.

Flow of the real-coded GA



Initial population for the example

Let the population size be $N = 8$.

Index	x_1	x_2
1	2.212	3.009
2	-2.289	-2.396
3	-2.933	-4.790
4	-0.639	1.692
5	-3.168	0.706
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Each row is already a real-valued chromosome. There is no binary string here.

How do we evaluate one solution?

Take solution 4:

$$\boldsymbol{x}^{(4)} = (-0.639,\ 1.692)^T$$

Substitute into the Rosenbrock function:

$$f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

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We repeat exactly the same process for all 8 solutions.

Evaluation of the full initial population

Index	x_1	x_2	$f(x_1, x_2)$
1	2.212	3.009	356.393
2	-2.289	-2.396	5840.936
3	-2.933	-4.790	17951.345
4	-0.639	1.692	167.469
5	-3.168	0.706	8722.680
6	0.215	-2.350	574.806
7	-0.742	1.934	194.424
8	-4.563	4.791	25726.938

Interpretation of the evaluation table

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At this stage, RGA behaves just like BGA from the selection point of view: evaluate, compare, and prefer better fitness values.

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Selection itself does not need any special real-coded modification. The major structural changes begin in crossover and mutation.

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This is exactly the same selection idea we studied in binary GA. Only the chromosome representation is different.

Binary tournament selection: step-by-step tournaments

Suppose we perform the following 8 tournaments.

Tournament	Candidates	Objective values	Winner	Reason
1	$(7, 6)$	(194.424, 574.806)	7	smaller value
2	$(4, 5)$	(167.469, 8722.680)	4	smaller value
3	$(8, 3)$	(25726.938, 17951.345)	3	smaller value
4	$(1, 2)$	(356.393, 5840.936)	1	smaller value
5	$(3, 1)$	(17951.345, 356.393)	1	smaller value
6	$(2, 8)$	(5840.936, 25726.938)	2	smaller value
7	$(7, 5)$	(194.424, 8722.680)	7	smaller value
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How the mating pool is formed from the tournaments

The winners of the 8 tournaments are:

$$7,\ 4,\ 3,\ 1,\ 1,\ 2,\ 7,\ 4$$

So the mating pool becomes:

New index	Original winner	(x_1, x_2)	$f(x_1, x_2)$
1	7	$(-0.742,\ 1.934)$	194.424
2	4	$(-0.639,\ 1.692)$	167.469
3	3	$(-2.933,\ -4.790)$	17951.345
4	1	$(2.212,\ 3.009)$	356.393
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Good solutions can appear multiple times in the mating pool. Weak solutions may disappear completely.

From the mating pool to crossover pairs

Now we randomly pair the solutions in the mating pool. Suppose the pair order is:

$$(3, 7), \qquad (5, 2), \qquad (8, 4), \qquad (6, 1)$$

- These are **mating-pool indices**, not original parent indices.

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This is one place where students often get confused. After selection, we usually relabel the mating pool from 1 to N and then perform pairing on these new positions.

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- At the same time, we want it to preserve useful genetic behavior.

A widely used real-valued crossover operator is **Simulated Binary Crossover (SBX)**.

Property 1 from binary GA: average is preserved

Consider a one-point crossover example in binary GA:

	Parent strings	Decoded value
Parent 1	101001	41
Parent 2	011010	26

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Average of offspring values:

$$\frac{42 + 25}{2} = 33.5$$

Property 1 from binary GA: average is preserved

Consider a one-point crossover example in binary GA:

	Parent strings	Decoded value
Parent 1	101001	41
Parent 2	011010	26

Average of parent values:

$$\frac{41 + 26}{2} = 33.5$$

After crossover, suppose the offspring are:

	Offspring strings	Decoded value
Offspring 1	101010	42
Offspring 2	011001	25

Average of offspring values:

$$\frac{42 + 25}{2} = 33.5$$

So the average before and after crossover is the same. This is one important behavior that SBX tries to mimic in continuous space.

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For this example,

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- $\beta < 1$: contracting crossover, offspring come closer together
- $\beta = 1$: stationary crossover, spread stays the same
- $\beta > 1$: expanding crossover, offspring move farther apart

Why these properties matter for SBX

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These two ideas - midpoint preservation and controlled spread - are the bridge from BGA crossover to real-coded crossover.

SBX formulas

For one variable, assume $p_1 < p_2$. SBX generates two offspring as:

$$o_1 = \bar{p} - \frac{1}{2}\beta(p_2 - p_1), \qquad o_2 = \bar{p} + \frac{1}{2}\beta(p_2 - p_1)$$

where

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- If $\beta = 1$, offspring spacing equals parent spacing.

How is the spread factor chosen in SBX?

The spread factor for the ith variable is sampled from a probability distribution controlled by η_c:

$$\beta_i = \begin{cases} (2u_i)^{\frac{1}{\eta_c+1}}, & u_i \leq 0.5 \\ \left(\frac{1}{2(1-u_i)}\right)^{\frac{1}{\eta_c+1}}, & u_i > 0.5 \end{cases}$$

where $u_i \in [0, 1]$ is a uniform random number.

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- This gives a balance between local search and exploration.

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Note: *SBX uses η_c to decide how conservative or adventurous the crossover should be.*

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Pair	Random number r
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- If $r \leq p_c$, crossover is applied.
- If $r > p_c$, the pair is copied unchanged.

Decision: which pairs actually crossover?

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This decision step is important. We do not jump directly from the mating pool to the offspring table.

Detailed SBX calculation for pair $(3, 7)$: variable x_1

From the mating pool:

$$\text{pair } (3,7): \quad (-2.933, -4.790)^T \text{ and } (-0.742, 1.934)^T$$

For variable x_1:

$$p_1 = -2.933, \qquad p_2 = -0.742$$

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So the two children from pair $(3, 7)$ are approximately $(-2.874, -4.773)^T$ and $(-0.801, 1.917)^T$.

Summary of the remaining crossover calculations

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Pair $(6, 1)$

- Using $\beta_1 \approx 0.914$ and $\beta_2 \approx 0.975$, the offspring are approximately:

$$(-2.222, -2.342)^T \quad \text{and} \quad (-0.809, 1.880)^T$$

Offspring population after crossover

Offspring	x_1	x_2	$f(x_1, x_2)$
1	-2.874	-4.773	17000.594
2	-0.801	1.917	165.908
3	-0.780	1.691	120.371
4	2.353	3.010	640.206
5	-0.639	1.692	167.469
6	2.212	3.009	356.393
7	-2.222	-2.342	5309.179
8	-0.809	1.880	153.462

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6	2.212	3.009	356.393
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Crossover can produce both better and worse solutions. Here offspring 3 and offspring 8 are already better than the best parent.

What students should notice after crossover

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This is why the selection operator and the variation operator play different roles: selection chooses promising parents, while crossover tries to create promising new combinations.

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where δ_i is generated from a polynomial-shaped distribution.

Mutation is usually applied with a small probability. It provides local perturbation and extra diversity.

How the mutation perturbation is generated

For polynomial mutation, one common expression is:

$$\delta_i = \begin{cases} (2r_i)^{\frac{1}{\eta_m+1}} - 1, & r_i < 0.5 \\ 1 - [2(1-r_i)]^{\frac{1}{\eta_m+1}}, & r_i \geq 0.5 \end{cases}$$

where $r_i \in [0, 1]$ is random and η_m is the mutation distribution index.

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This is why polynomial mutation is useful: it usually performs local refinement, but it still keeps a chance of making a larger move.

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- Therefore η_m controls the aggressiveness of mutation.

Note: *SBX controls how offspring are placed between or beyond parents, while polynomial mutation controls how strongly a single solution is perturbed.*

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Therefore:

- offspring 2 mutates
- offspring 4 mutates
- offspring 8 mutates
- the other offspring are copied unchanged

Detailed mutation calculation: offspring 2

Before mutation, offspring 2 is:

$$(-0.801,\ 1.917)^T$$

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So the mutated offspring becomes approximately $(-0.528, 0.564)^T$, which is a very strong improvement.

Brief mutation examples for other selected offspring

Offspring 4 before mutation:

$$(2.353,\ 3.010)^T$$

Using suitable random numbers, suppose it mutates to:

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This comparison is useful in class: mutation can improve a solution strongly, improve it slightly, or even make it worse.

Offspring population after mutation

Offspring	x_1	x_2	$f(x_1, x_2)$
1	-2.874	-4.773	17000.594
2	-0.528	0.564	10.470
3	-0.780	1.691	120.371
4	1.969	3.140	55.250
5	-0.639	1.692	167.469
6	2.212	3.009	356.393
7	-2.222	-2.342	5309.179
8	0.281	1.850	314.175

Summary after mutation

- offspring 2 improves dramatically after mutation

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Note: mutation is not guaranteed to improve a solution. Its purpose is to inject perturbation and diversity. Selection later decides which solutions deserve to survive.

Survivor selection: $(\mu + \lambda)$ strategy

To form the next generation, we combine:

- the parent population of size $\mu = 8$

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Then we sort all 16 solutions by objective value and keep the best 8.

This is called the $(\mu + \lambda)$ strategy. Strong parents can survive, and strong offspring can replace weak parents.

Survivor selection: ranking the combined population

The best solutions after combining parents and offspring are:

Rank	Source	Index	(x_1, x_2)	Objective
1	Offspring	2	$(-0.528, 0.564)$	10.470
2	Offspring	4	$(1.969, 3.140)$	55.250
3	Offspring	3	$(-0.780, 1.691)$	120.371
4	Parent	4	$(-0.639, 1.692)$	167.469
5	Offspring	5	$(-0.639, 1.692)$	167.469
6	Parent	7	$(-0.742, 1.934)$	194.424
7	Offspring	8	$(0.281, 1.850)$	314.175
8	Parent	1	$(2.212, 3.009)$	356.393

Next-generation parent population

So the next parent population becomes:

New index	x_1	x_2	$f(x_1, x_2)$
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This becomes the parent population for generation $t = 2$.

What changed and what stayed the same from BGA to RGA?

Aspect	BGA	RGA
Representation	binary strings	real-valued vectors
Selection	based on fitness	based on fitness
Crossover	bit-level crossover	SBX or other real-valued crossover
Mutation	bit flip	polynomial or other real perturbation
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This comparison is worth emphasizing: selection logic remains the same, but representation and variation operators are redesigned.

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- Why can mutation both improve and worsen a solution?

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- A. fixed binary strings only

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